AarogyaTwin-X

- Forecasts with evidence, not unsupported certainty
- Local synthetic patient replay and glucose forecasting
- Not a diagnostic device. Not medical advice.

The question

- Can a forecast be useful without hiding its limitations?
- Users need data freshness, uncertainty, and provenance
- Missing data should trigger abstention, not invented certainty

The experience

- Select a held-out synthetic patient
- Replay the timeline in five-minute steps
- Inspect context, forecast heads, intervals, event risk, and evidence

Synthetic data only

- 60 artificial patients x 5 days x 5-minute observations
- Static records plus CGM, heart rate, HRV, sleep, activity, and meals
- Seeded variability, rhythms, noise, and missingness
- No real patient data

Leakage-aware evaluation

- 36 training patients
- 6 calibration-tuning patients
- 6 calibration-interval patients
- 12 held-out test patients
- Past-only features at every replay anchor

Model architecture

- Small CPU temporal lag gradient-boosting heads
- Calibrated fusion logistic event classifier
- Train-only preprocessing and missingness indicators
- Conformal intervals and explicit ablations

Personalization

- Make a forecast at time T
- Wait for the target timestamp
- Use a bounded history of matured errors for later forecasts
- Never use the current forecast’s future target
- Adaptive residual, not clinical physiology

Held-out forecast results

- 30 min: MAE 7.21 | RMSE 9.76 | coverage 89.95%
- 60 min: MAE 7.85 | RMSE 10.42 | coverage 89.36%
- 120 min: MAE 8.38 | RMSE 11.00 | coverage 90.00%
- Intervals are calibrated named heads, not extra points

Probability calibration

- Fusion Brier: 0.05035 -> 0.02437
- Fusion log loss: 0.18466 -> 0.13559
- Expected calibration error: 0.05814 -> 0.02387
- AUROC 0.94330 and AUPRC 0.27652 remained ranking measures

Missing data

- Natural missingness: about 0.3% abstention
- Recent stream gaps degrade measured forecasts
- Recent 30-minute CGM dropout: 100% abstention
- Unavailable estimates remain null

Distribution shift

- Nominal 120-minute coverage: 89.46%
- Glucose level +25 mg/dL: coverage 62.09%
- Meal inputs doubled: coverage 89.35%
- HbA1c +1.0: coverage 80.26%
- Stress tests show known failure modes

Alert behavior

- 22 synthetic episodes across 12 test patients
- At threshold 0.10: episode sensitivity 22.7%, median lead 70 min
- At threshold 0.20: episode sensitivity 18.2%, 0.133 false alerts/patient-day
- Synthetic alert metrics are not clinical performance

Live demo

- Show synthetic/non-clinical warning
- Select held-out record and replay +5 minutes
- Inspect heads and intervals
- Enable dropout and show abstention
- Review scenario disclaimer and measured evidence

Safety and transparency

- No diagnosis, dosing, medication change, or treatment advice
- Drivers are observed context, not causal attribution
- Scenario controls are assumptions, not treatment effects
- Not-ready, loading, error, and unavailable states are explicit

Reproducibility

- Seed 2026 and local synthetic generation
- Joblib bundle plus strict JSON metrics/provenance
- Engine loads artifacts and never trains automatically
- API and dashboard run on localhost

Limitations

- One simulator and one seeded cohort
- Correlated replay windows
- Small event count for alert conclusions
- Coverage can fail under distribution shift
- No clinical validation

Closing

- A reproducible foundation for honest forecasting research
- Past-only temporal features
- Measured uncertainty and probability calibration
- Data-quality abstention and explicit limitations